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SUPERSEDING
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INTERIM FEDERAL SPECIFICATION

CLOTH, COTTON, MUSLIN (MERCERIZED)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for mercerized cloth, cotton, muslin.

1.2 Classification

1.2.1 Type. Cloth covered by this specification shall be of one type.

1.2.2 Colors. Unless otherwise specified, the cloth shall be of the following solid colors, as specified (see 3.4.1, and 6.2):

White	Red
Blue	Rose
Purplish Red	Turquoise
Green	Yellow

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 191 - Textile Test Methods.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 8305

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 303 - Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

Color Association of the United States, Inc.

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Ave., New York, NY 10016.)

3. REQUIREMENTS

3.1 Preproduction sample approval. When specified in the contract or order, before production is commenced, a 1/2 yard sample of cotton muslin cloth shall be submitted to the contracting officer for approval (see 6.2).

3.2 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and finish and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Material.

3.3.1 Yarn. The yarn shall be made from cotton which has been cleaned, carded, drawn and spun into singles yarn for both warp and filling.

3.4 Color. The cloth shall be bleached white or dyed. The use of sulfur dyes and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.5.

3.4.1 Matching. Unless otherwise specified, and no standard sample of the cloth is available, colors as specified in 1.2.2 shall approximate the following numbers of Fed. Std. No. 595 and cable numbers of the Standard Color Card of America.

Fed. Std. No. 595	
Color	Number
Blue	15123
Purplish Red	21158
Green	24260
Yellow	33695

Standard Color Card of America	
Color	Cable Number
White	70001
Red	70180
Rose	70099
Turquoise	70021

3.4.1.1 Colors shall be matched as specified in 3.4.1, or colors other than specified in 1.2.2 when required shall be matched to the approved preproduction sample or standard sample, under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the sample under incandescent lamplight at 2800° Kelvin.

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3.5 Physical requirements. The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.5.

3.5.1 Width. Unless otherwise specified, the width of cloth shall be 35 to 36 inches (see 6.2).

3.5.2 Length. Unless otherwise specified, the length of cloth shall be 30 to 35 yards in one continuous piece, put up in flat folded pieces or bolt form (see 6.2).

TABLE I. Physical requirements

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (grab method) (minimum)		Weave
Ounces	Warp	Filling	Warp	Filling	
5.0	53	46	60	60	Plain

3.6 Finish. The cloth shall be mercerized and have a crisp, lustrous finish to match the standard sample and shall show no appreciable change in appearance or hand after three launderings when tested as specified in 4.5.

3.6.1 Nonfibrous material. Total sizing, finishing and other nonfibrous material content of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.5.

3.7 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 1.0 percent in either the direction of the warp or the filling when tested as specified in 4.5. The preshrinking process used shall not be identified by name or trademark, either on the cloth, the ticket, or package.

3.8 Tickets. Each piece shall have an identification ticket attached as specified in PPP-P-1134.

3.9 Fiber identification. Each bolt or flat folded piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship. The cloth shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the quality level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Preproduction sample inspection. When a preproduction sample is required, inspection shall be made of the finished sheet for conformance with all provisions of this specification.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing and marking comply with section 5 of this specification.

4.3.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 Yard-by-yard examination. The required yardage of each bolt or flat folded piece shall be examined on the face side and all defects as defined in Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.2.1, except that spots, stains and streaks shall be counted regardless of cause (i.e. not restricted to rust, oil, or grease) and only those slubs and knots which exceed the maximum acceptable limits shown for each shall be scored see 4.3.2.2). Defects on the selvage

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shall not be scored unless the body of the cloth is affected. o yard shall be penalized more than four points. For purpose of this examination, a yard shall begin at the last scored defect. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined in 25 bolts or flat folded pieces selected from 25 containers. An approximate equal number of yards shall be examined in each bolt or flat folded piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceed the following value:

28.00 points

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of the material}} = \text{Points per 100 square yards.}$$

4.3.2.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

Note: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Objectionable odor.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.
Excessive neppiness.
Finish does not match standard sample.

4.3.2.2 Limits of slubs and knots. Only slubs and knots which exceed the limits shown on figure 1 shall be scored.

4.3.3 Examination for length.

4.3.3.1 Individual bolts or flat folded pieces. During the yard-by-yard examination, each bolt or flat folded piece shall be examined for gross length. Any gross length found to be less than the minimum specified, or more than two yards below the gross length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more bolts or flat folded pieces in the sample are defective in respect to length.

4.3.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual gross lengths of bolts or flat folded pieces in the sample is less than the total of the gross lengths marked on the tickets.

4.3.4 Examination for shade. During the yard-by-yard examination, each bolt or flat folded piece shall be examined for shade. Any bolt or flat folded piece in the sample, off shade (compared to the applicable standard), shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.3.5 Examination for identification of preshrinkage process and noncompliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each bolt or flat folded piece shall be examined for these defects. The lot shall be unacceptable if two or more bolts or flat folded pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

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4.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

4.5 Testing of the end item. The methods of testing specified in Fed. Std. 191, wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous yards full width of the finished cloth. The sample size shall be as shown in the table below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

TABLE II. Tests

Characteristic	Requirement reference	Test method
Weight	Table I	5041
Yarns per in	Table I	5050
Weave 1/	Table I	Visual
Breaking strength - pounds	Table I	5100.1
Color fastness to:		
Light 2/	3.3.2	5660.1
Laundering	3.3.2	5610.1
Perspiration 3/	3.3.2	5680
Crocking	3.3.2	5651
Identification of mercerization 1/ 4/	3.5	Microscopic examination for swelling of cotton fiber.
Sulfur content 1/	3.4	4.5.1
Nonfibrous material content	3.5	2611
Shrinkage in laundering 5/	3.6	5550.1
Finish 6/	3.5	4.5.2

1/ One determination shall be made from each sample unit and the results reported as "Pass" or "Fail".

2/ In the event of a dispute resulting from the test with method 5660.1 or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.2.

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3/ The transfer cloth shall consist of a warp of polyester fiber or Orlon 81 and a 6-fiber filling, with equal bars of wool, spun nylon, spun rayon, spun silk, spun acetate and bleached cotton. Each 6-fiber repeat shall measure approximately 2 inches and shall be separated by a waste filling stripe.

4/ A contractor's certificate of compliance shall be acceptable for this characteristic.

5/ The specimens shall be selected from the cloth (sample unit) as follows: One (1) specimen from each side of the cloth to within 3 inches of selvage and one (1) specimen from the center of the cloth. No two specimens shall contain the same filling yarns. Each elongation value shall be prefixed with a minus sign and both inclosed in parenthesis.

6/ One determination shall be made after three launderings from each sample unit and the results reported as "Pass" or "Fail".

4.5.1 Test for sulfur content.

4.5.1.1 Solution (acid stannous chloride). Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.5.1.2 Procedure. One to 1.5 grams of the finished cloth, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with stannous chloride solution. Place a filter paper which has been moistened with one drop of 5.0 percent lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.5.1.3 Evaluation. The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory. When no standard sample is available, the sample shall be considered satisfactory if there is no more than a slight discoloration in the lead acetate test and with no loss in breaking strength above that of a vat dyed cotton specimen of equal construction and weight.

4.5.2 Test for launderability of finish. The test specimen shall be two 10 inch by 10 inch pieces of the finished cloth (specimens A and B). Specimen A shall be kept for comparison purposes and shall not be laundered. Specimen B shall be subjected to 3 complete launderings in accordance with the method of laundering specified in test method 5550.1. After the launderings, specimen B shall be compared to specimen A for appearance and hand.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The packaging, packing and marking shall be in accordance with PPP-P-1134. The level of packaging shall be A, B, or C as specified (see 6.2), the level of packing shall be A, B, or C as specified (see 6.2) and the marking shall be civil or military as specified (see 6.2).

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use as dress goods and suiting material.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

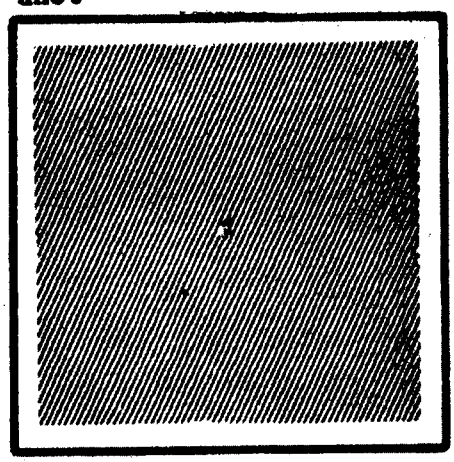
(a) Title, number, and date of this specification.
(b) When preproduction sample is required (see 3.1).
(c) Color of cloth required (see 3.4 and 3.4.1).
(d) When different width is required (see 3.5.1).
(e) When different length is required (see 3.5.2).
(f) Selection of applicable levels of packaging and packing (see 5.1).

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6.3 Stand sample. For access to standard sample, address the procuring office issuing the invitation for bids.

Limits of Acceptability

Knot



Yarn Thickening (Slub)

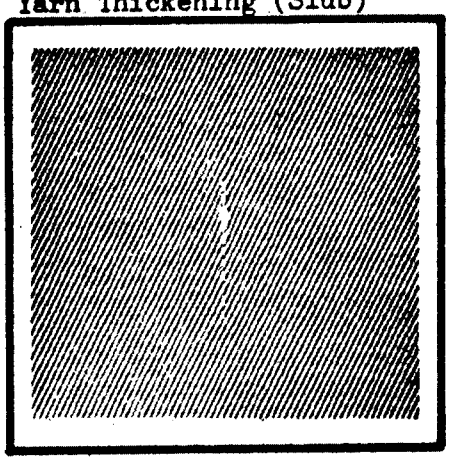


Fig. 1

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